USB

File: USB.kicad_sch

Power

File: Power.kicad_sch

MCU

File: MCU.kicad_sch

RX_Frontend

File: RX_Frontend.kicad_sch

TX_PA

File: TX_PA.kicad_sch

CW_Synth

File: CW_Synth.kicad_sch

BEACON Phone Module — inferred specifications (no spec doc supplied; derived from the Altium schematic)
  * Plugs straight into the phone through a USB-C plug: USB 2.0 FS data + 5 V power (<= 500 mA).
  * Illuminates the backscatter wearable with a 2.4 GHz ISM CW carrier: LMX2582 -> CC2595 PA, about +20 dBm
    conducted into a U.FL for a Taoglas FXP70.07.0053A flex antenna, low-pass filtered for harmonics.
  * Receives the backscattered signal on a Taoglas WDP.2458 ceramic patch: BPF -> QPL9547 LNA -> nRF52840.
  * Housed in a MagSafe puck (beacon_phone_magsafe_2p5mm.scad): 55 x 78.45 mm board, 4x M2, USB-C cable to the phone.
  * nRF52840 controls the synth (SPI), PA_EN, LNA shutdown and the USB link; it reads two INA219 monitors
    (5 V input and 3.3 V rail).
  * 3.3 V rail from an LT80602 Silent Switcher buck; expected load about 0.45 A.
Top-sheet TODOs from the Altium file: RX/TX antenna connectors added; PA GND connected; MCU and CW synth completed.

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HW: M. Marsicano, J. Oshinsky, N. McKillips, I. Hong | FW: J. Faubus, J. Hensley
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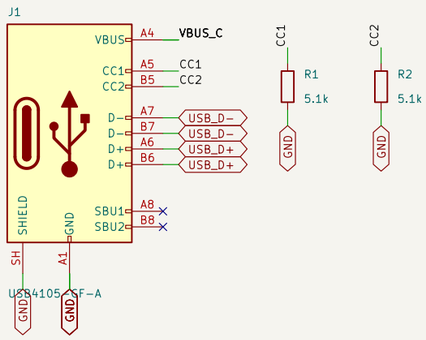
Sheet: /
File: BEACON_PhoneModule.kicad_sch

Title: BEACON Phone Module

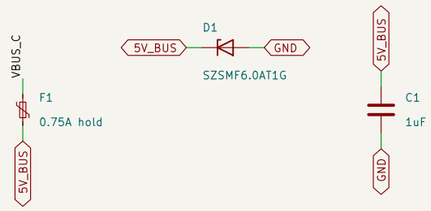
Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/7

USB sheet notes
- MagSafe puck: a USB-C receptacle at the bottom edge takes a short C-to-C cable to the phone (the phone is host and 5 V source). Rd = 5.1k on both CC1 and CC2. (The Altium drawing used a UP20 plug for the direct-plug dongle concept.)
- USB 2.0 full-speed device: D+/D- run straight to the nRF52840 (internal series R).
- Keep total VBUS capacitance <= 10 uF (USB 2.0 inrush).

USB-C receptacle (short C-C cable to the phone) - Rd 5.1k on CC1 and CC2 advertises a sink



VBUS protection: polyfuse + 6.0 V TVS



USB D+/D- ESD (USBLC6: pins 1/6 and 3/4 are one line each - flow-through)

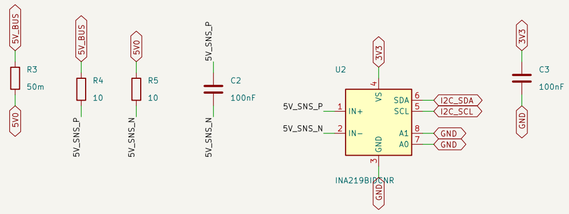
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Sheet: /USB/
File: USB.kicad_sch

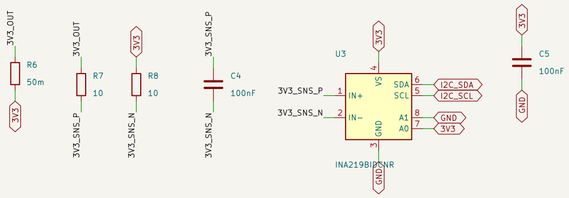
Title: USB-C receptacle, VBUS protection and ESD

Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 2/7

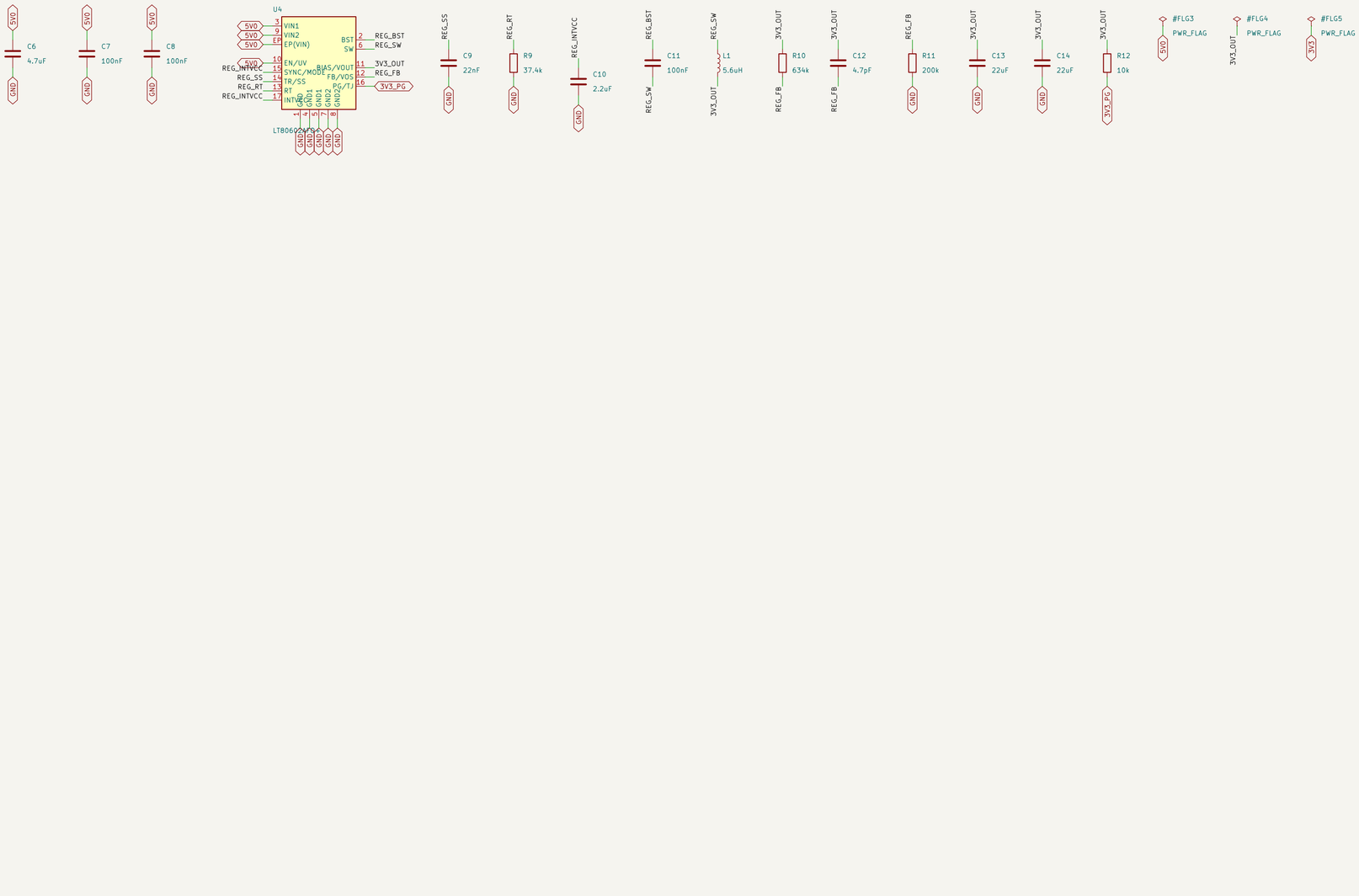
5V INA219 current/voltage monitor — shunt 5V_BUS -> 5V0, I2C addr 0x40



3V3 INA219 current/voltage monitor — shunt 3V3_OUT -> 3V3, I2C addr 0x41



3.3 V Silent Switcher buck (LT80602): 5V0 -> 3V3_OUT, forced-continuous at a fixed ~506 kHz



Power sheet notes
- CHANGE: the buck now runs from 5V0 (after the 5 V shunt) so the 5 V INA219 measures the whole module input. In the Altium drawing the buck tapped 5V_BUS ahead of the shunt.
- 5V0 also feeds the QPL9547 LNA (its recommended supply is 3.15–5.25 V, 5 V typ).
- CHANGE: shunts 5 mOhm -> 50 mOhm. Expected 3V3 load ~0.45 A (LMX2582 ~0.25 A, CC2595 ~0.1 A, nRF52840 + misc), so 5 mOhm only used 1/16th of the ADC range.
- INA219 addresses: 5V monitor 0x40 (A0=A1=GND), 3V3 monitor 0x41 (A0=VS).
- FIX: FB divider 681k/220k -> 634k/200k (ADI: RFB−BOT <= 205k). Vout = 0.8 x (1+634/200) = 3.34 V, leaving headroom for shunt + bead drops to the LMX2582 (3.15 V min).
- RT 37.4k -> fSW ~506 kHz (ADI eq. 2); L = 0.75 x Vout/fSW = 4.9 uH -> 5.6 uH: CBST 1 uF (ADI Table 7). Pinout verified against ADI datasheet Rev 2, Table 3.
- LT80602 exposed pad is internally VIN: it connects to 5V0, NOT GND.
- CHANGE: SYNC/MODE GND (Burst) -> INTVCC (forced continuous): fixed 506 kHz keeps the switching spurs at a known offset from the carrier; BST cap is 0.1 uF in FCM.

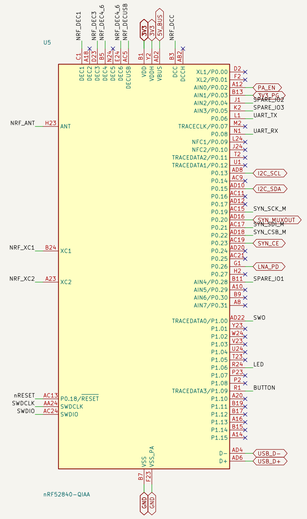
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HW: M. Marsicano, J. Oshinsky, N. McKillips, I. Hong
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Sheet: /Power/
File: Power.kicad_sch

Title: Power: current monitors and 3.3 V buck

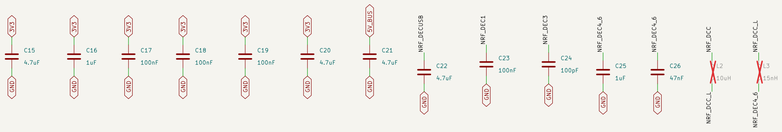
Size: A2	Date: 2026-11-02	**Rev: A**
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nRF52840-QIAA (aQFN73)

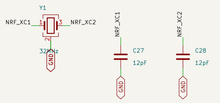


MCU sheet notes:
- Supply: VDD and VDDH both on 3V3 (normal-voltage mode); DCCH unused, REG1 runs in LDO mode (firmware DCDCEN=0); the 10 uH ferrite would saturate in the MagSafe magnet field, so the DC/DC inductors are DNP footprints. VBUS from USB for the USBD peripheral.
- Removed wearable-only nets copied from the wearable MCU sheet: CTRL_SPDT, Mod_Sel0/1, PMIC_Status0-2, Sensor_SDA/SCL/INT/EN, QSPI_CLK.
- GPIO map: SYN_SCK P0.19, SYN_MUXOUT P0.20, SYN_SDI P0.21, SYN_CSB P0.22, SYN_CE P0.23, I2C_SCL P0.13 / SDA P0.15, PA_EN P0.02, LNA_PD P0.26, 3V3_PG P0.03, LED P1.06, BUTTON P1.09, UART TX P0.06 / RX P0.08, spares P0.28 / P0.04 / P0.05, SWD P1.00.
- P0.02, P0.04, P0.28 and P1.06 are radio-sensitive (<=10 kHz, standard drive); used as GPIO for static enables, PA and the LED.
- No 32.768 kHz crystal: use the internal LFRC (XL1/XL2 left open).

Supply decoupling (Nordic PS v1.11 reference circuitry)



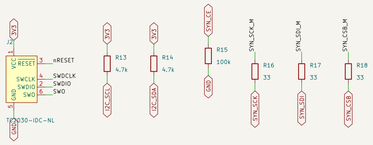
32 MHz crystal (CL = 8 pF -> 12 pF load caps)



ANT matching (Nordic v1.1: 0.8 pF / 4.7 nH / 0.5 pF) — RX_RF comes from the LNA



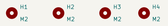
SWD (Tag-Connect TC2030-IDC-NL), I2C pull-ups, synth CE default-off



User LED and demo button



M2 mounting holes (enclosure heat-set inserts; 6 mm keepout both sides)



Test points



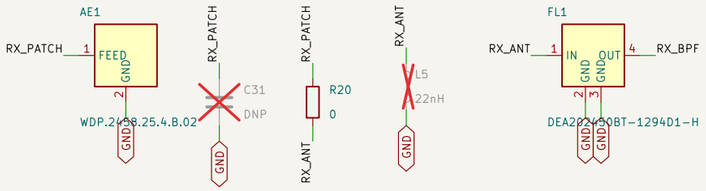
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Sheet: /MCU/
File: MCU.kicad_sch

Title: nRF52840 MCU / BLE receiver, debug, UI

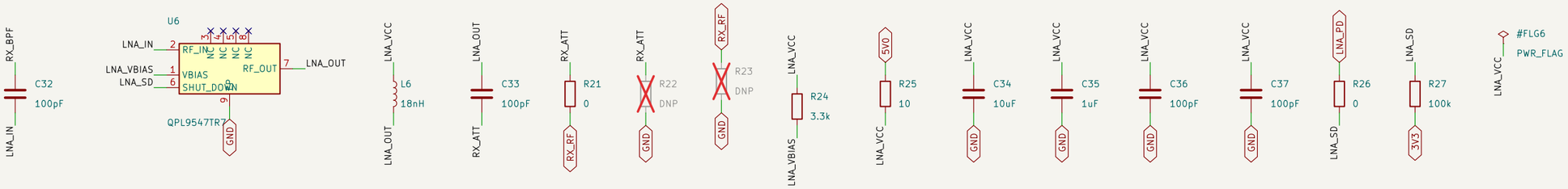
Size: A1	Date: 2026-11-02	Rev: A
KiCad E.D.A. 10.0.7		Id: 4/7

RX notes
- RX antenna: Taoglas WDP.2458.25.4 25x25 mm pin-fed patch at the board centre (MagSafe enclosure).
  Feed runs on B.Cu from the pin, then an RF via to L1. Pi match footprints for tuning.
- The LNA now runs from 5V0 (QPL9547 recommended 3.15-5.25 V).
- LNA is OFF at reset (100k pull-up on Shut_Down); firmware drives LNA_PD low to enable.
- LNA supply through a ferrite bead from 5V0 (keeps buck ripple off the LNA).
- RX OVERLOAD: TX carrier leaking into the RX antenna is in-band and gets ~17 dB of gain.
  Keep antenna-to-antenna isolation >= 30 dB, or fit the pi pad (R_ATT shunts) so the
  nRF ANT pin stays below 0 dBm (abs max +10 dBm). The nRF must never transmit: its ANT
  pin drives straight into the LNA output.
- All RF lines: 50 Ohm grounded CPW, see the PCB stackup notes.

RX antenna: Taoglas WDP.2458.25.4 ceramic patch (pin-fed, board ground = its ground plane) + pi match



QPL9547 LNA (Qorvo EVB values; ~17.8 dB gain at 2.4 GHz)



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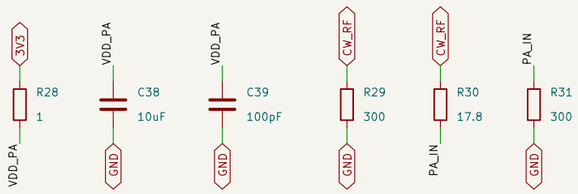
Sheet: /RX_Frontend/
File: RX_Frontend.kicad_sch

Title: RX chain: antenna, BPF, QPL9547 LNA

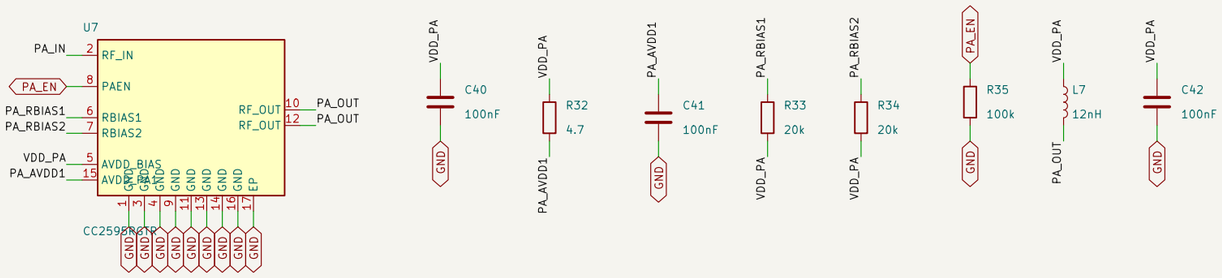
Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/7

TX notes
- FIX: CC2595 RF_OUT (pins 10/12) is the PA collector and needs a DC feed: 12 nH from VDD_PA as on the TI CC2595EM. It was missing from the Altium drawing.
- FIX: RBIAS1/RBIAS2 20k go to VDD_PA per the TI EM schematic (the drawing tied them to GND). The TI BOM table conflicts with this; confirm on first boards.
- GND on every CC2595 GND pin and the EP (TODO "add GND connection to PA").
- TX antenna: Taoglas FXP70.07.0053A flex antenna on a U.FL at the right edge (53 mm cable).
- PA input pad: the LMX2582 gives several dBm; 3 dB pad plus OUTA_PWR register sets ~0 dBm into the PA (abs max +10 dBm).

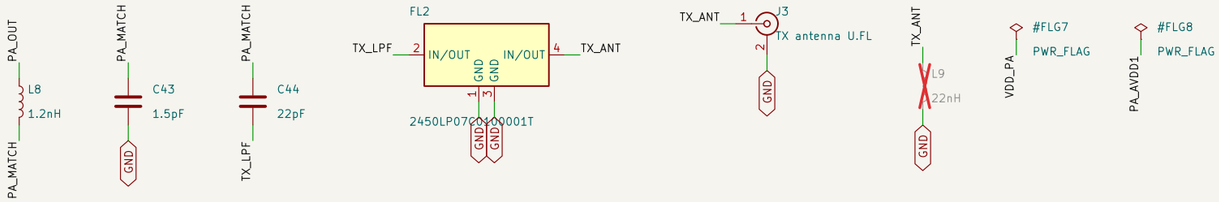
PA supply (filtered 3V3) and 3 dB input pad (sets PA drive ~0 dBm)



CC2595 PA (TI EVM values) — +20.7 dBm typ at 0 dBm in



Output match (1.2 nH / 1.5 pF), DC block, 2.45 GHz low-pass, TX antenna connector



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Sheet: /TX_PA/
File: TX_PA.kicad_sch

Title: TX chain: pad, CC2595 PA, LPF, antenna

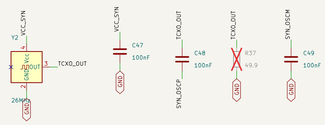
Size: A3	Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 6/7

CW synthesizer notes (completed from the TI typical application)
- Carrier: 2400–2483.5 MHz ISM, VCO 3.55–7.1 GHz, CHDIV=2 -> 2.4 GHz on OUTA.
- Reference: 26 MHz TCXO. The PFD can use the OSCin doubler (52 MHz) in fractional mode.
- SPI: SCK/SDI/CSB from the nRF52840; MUXout = lock detect or SPI readback.
- CE has a 100k pull-down on the MCU sheet: the synth stays off until firmware enables it.
- OUTB powered down in firmware: pins left open.
- Loop filter: PFD 52 MHz (OSC_2X), Icp 7.27 mA -> LBW ~59 kHz, PM ~62 deg. A wider loop is not reachable with CN >= 3.3 nF and would worsen 1–5 MHz phase noise (the tag band).
- Spur-free channels with a 26 MHz TCXO: multiples of 13 MHz (2405, 2418, ... 2483 MHz).
- Register plan and per-channel words: docs/lmx2582_loop_filter.md (verify with TICS Pro).

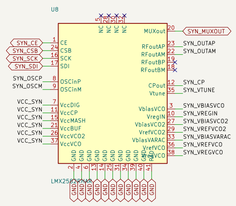
Filtered 3.3 V for the synthesizer (3.15–3.45 V)



26 MHz TCXO reference (clipped sine), AC-coupled into OSCin



LMX2582 PLL + VCO (TI SNAS680E Fig 8-11)



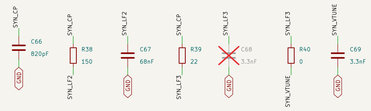
Supply decoupling: 0.1 uF on every Vcc pin, +10 uF on VccMASH / VccVCO / VccVCO2



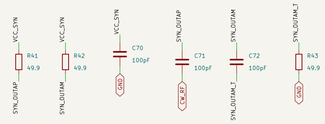
Internal regulator / VCO bias bypass caps (place at the pins)



Loop filter: PFD 52 MHz, Icp 7.27 mA, LBW ~59 kHz, PM ~62 deg (docs/lmx2582_loop_filter.md)



RF output A: 50 Ohm pull-ups, 100 pF DC blocks; OUTAM terminated in 50 Ohm (B unused)



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Sheet: /CW_Synth/
File: CW_Synth.kicad_sch

Title: CW synthesizer: LMX2582 + TCXO	
Size: A1 | Date: 2026-11-02	**Rev: A**
KiCad E.D.A. 10.0.7	Id: 7/7